Dhairya Shah

Gujarat, India | dhairyagshah233@gmail.com
linkedin.com/in/dhairyagshah233 | github.com/Dhairya233 | dhairya233.github.io

EDUCATION

Ahmedabad University Class of 2028
B.Tech, Computer Science and Engineering (3rd year) CGPA: 3.25/4.0

PROJECTS

GestureOS | Python, MediaPipe, OpenCV, CustomTkinter github.com/Dhairya233/GestureOS

Control a Windows PC with your bare hands, through a webcam.

- **Built a Windows desktop app** that turns webcam hand tracking (MediaPipe Hand Landmarker, 21 landmarks per hand) into mouse, scroll, media, drawing and app-launch controls.
- **Designed a safety model:** the app starts disarmed, arms by a two-palm hold, disarms from any window with a global hotkey, and sends every OS event through one gateway that releases held keys on disarm.
- **Wrote 630 automated tests** that drive the real engine with generated hands and a fake input backend, so the project is tested without a camera, with CI on GitHub Actions.

Life-RPG Protocol | Next.js, TypeScript, Supabase/PostgreSQL github.com/Dhairya233/life-rpg-protocol

A gamified productivity app where I found, then fixed, a privilege-escalation hole in my own database.

- **Built a gamified productivity web app** (Next.js 14, TypeScript, Supabase/PostgreSQL) with a focus-session monitor and reputation-weighted peer review of quest proof photos.
- **Audited my own database and found users could edit their own Aura, XP and role**; fixed it with column-level grants, revoked function rights and identity taken from auth.uid().
- **Wrote the attacks as tests:** 44 attack checks failed on the old schema and about 260 database checks now pass in CI, including a two-session race test against double payouts.

Browser Arcade | JavaScript, WebRTC, PeerJS github.com/Dhairya233/browser-arcade

Two multiplayer board games where I found, then fixed, a hidden-information leak in my own game.

- **Built two multiplayer browser board games, Shadow Hunt and Game of Trades,** in plain JavaScript modules with WebRTC (PeerJS) and no build step.
- **Found a hidden-information leak in my own game** (26 of 26 snapshots exposed the hidden player's position) and fixed it with a host-authoritative design: per-seat views and validation of every action before it is applied.
- **Wrote a test that reads every message sent to every non-hidden seat across a whole online game**, plus a fuzz test with hostile payloads; 135 tests run on every push with GitHub Actions.

Fabric Defect Anomaly Detection | Python, scikit-learn github.com/Dhairya233/Textile-Defect-ML-Model

A leak-aware study of how much your data split changes a fabric-defect detector's results.

- **Built a solo, leak-aware study of classical fabric-defect detection on the public AITEX dataset** (245 strips, 7 fabrics): 90 texture features per patch feeding a random forest, an SVM and four anomaly detectors, with a static results explorer.
- **Measured the cost of a careless split:** a random patch split overstated the random forest's patch-level average precision by 0.44 (95% interval 0.27 to 0.53) against splitting by whole strip, and by 0.64 when a whole fabric was held out; with a fabric held out every method scored close to chance.
- **Resampled whole strips (1,000 bootstrap resamples)** for intervals and wrote 75 passing tests, including checks that no strip sits on both sides of a split and that the numbers quoted in the README match the result tables.

RESEARCH EXPERIENCE

Institute of Manufacturing and Economy, Ahmedabad University Snapshot published August 2026
Research Intern ahduni.edu.in (Institute Insights: textiles sector snapshot)

- **Conducted the primary research** for the Institute's published sector snapshot "Textiles: Growing in Scale, Losing Relative Ground" (NIC Divisions 13 and 14, August 2026).
- Worked from **the Annual Survey of Industries, the Index of Industrial Production, DGCI&S trade statistics and CMIE Prowess** firm-level data, covering 2014-15 to 2024-25.
- **Findings include 61 percent growth in sector output from 2014-15 to 2023-24** alongside a lag behind overall manufacturing on the production index, and imports growing faster than exports.

SKILLS

Languages Python JavaScript TypeScript SQL
Tools Next.js Supabase/PostgreSQL WebRTC MediaPipe OpenCV scikit-learn pytest GitHub Actions Git
Learning Data structures and algorithms C++ Data engineering
Spoken languages English Hindi Gujarati **Hobbies** Music

CERTIFICATIONS

AWS Academy Cloud Foundations, AWS Academy
AWS Academy Graduate - Data Engineering, AWS Academy
Certificates available at dhairya233.github.io (Documents)